

An Analysis of the Correlation Between Gold and S&P 500 Returns (2000-2025)

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Abstract

This paper examines the correlation between gold and S&P 500 monthly returns across five periods from January 2000 to December 2025, including the full sample and four distinct market regimes, comprising 312 monthly observations. Pearson's correlation coefficient is used for the full sample, where the large number of observations supports the assumptions required for the test. For the four crisis sub-periods, where sample sizes range from 24 to 46 observations, Spearman's rank correlation is utilized since smaller samples are more susceptible to non-normality and outliers. Results show no statistically significant correlation in any period examined under either method. For the full sample, the near-zero correlation suggests that gold exhibited little to no average relationship with the S&P 500, behavior that is consistent with a weak hedge under the framework of Baur and Lucey (Baur and Lucey 2010, 219). However, the direction of the sub-period coefficients varies meaningfully across crisis types: correlations are negative during the Dot-Com Crash and the Great Recession, suggesting some consistency with safe-haven behavior, and positive during the COVID-19 period, reflecting a period in which both assets declined simultaneously. These findings suggest that while gold may function as a weak hedge on average, its role during periods of market stress depends on the nature of the crisis and cannot be assumed to remain consistent across all market environments.

I. Introduction

Gold is often marketed to investors as a portfolio hedge when the market declines. The conventional narrative holds that gold and stocks move inversely: when equity markets fall, investors turn to gold, driving its price higher. This relationship supports the allocation of gold in diversified portfolios and underlies many retail and institutional investment strategies.

Yet the evidence supporting this claim is mixed. The relationship between gold and equities is not static. It appears to change depending on the market crisis, the monetary policy environment, and investor behavior during such moments where markets are tested. For instance, a crisis

caused by a lack of liquidity can affect gold and stocks differently than a crisis driven by a credit collapse or a geopolitical event.

This paper looks to find the correlation between gold and equities by using monthly return data for gold and the S&P 500 from January 2000 to December 2025. Rather than only examining a single, uninterrupted time period, the analysis is structured around five distinct periods: a full-sample baseline and four sub-periods selected to capture different types of market stress. The primary research objective is to determine gold's behavior both on average and during periods of market turmoil.

II. Conceptual Framework: Hedge, Diversifier, and Safe Haven

To interpret the study's results, it is important to have an understanding of what it means for gold to be a hedge or a safe haven. Dirk G. Baur and Brian M. Lucey provide one of the more widely adopted definitions in the finance literature, distinguishing between three related but distinct concepts: hedge, safe-haven, and diversifier (Baur and Lucey 2010, 219).

A hedge is defined as an asset that is uncorrelated or negatively correlated with another asset or portfolio on average. A crucial component of a hedge is the fact that it does not carry the property of reducing losses in times of market stress or turmoil, as the asset could, for instance, exhibit a positive correlation in such periods and a negative correlation in normal times, with a negative correlation on average.

A diversifier is an asset that is positively, but not perfectly (correlation coefficient would be 1), correlated with another asset or portfolio on average. Like a hedge, a diversifier does not necessarily provide protection in extreme adverse market conditions, since a diversifier only observes the correlation on average.

A safe haven is defined as an asset that is uncorrelated or negatively correlated with another asset or portfolio in times of market stress or turmoil specifically. The safe haven property does not require a particular correlation in normal market conditions. Rather, the correlation of a safe haven is negative during extreme adverse events. If the safe haven asset is negatively correlated with another asset during such periods, it actively compensates investors for losses because its price rises as the other asset falls.

III. Data and Methodology

A. Data

Gold prices are collected from USAGOLD, a source that compiles daily gold XAU/USD prices. S&P 500 indices are drawn Macrotrends, which is another source that tracks historical market records. The price on the last trading day of the month is recorded each month. The sample covers 312 monthly observations from January 2000 to December 2025.

All analysis is conducted on monthly returns rather than price levels. Monthly return is defined as the percentage change from the prior month-end closing price to the current month-end closing price. This approach is methodologically preferable to using price levels because both gold and stock prices naturally climb over time. Comparing two naturally rising numbers may create a false correlation that does not truly reflect the relationship between the two.

B. Sub-Period Definitions

To test whether gold behaves as a safe haven in specific periods of market stress, the sample is divided into four crisis sub-periods in addition to the full-sample baseline. The Great Recession period is defined starting in June 2006 rather than at the formal beginning of the Global Financial Crisis in late 2007. This earlier start date reflects the collapse of the U.S. housing market that preceded and directly caused the broader financial crisis. This also increases the sample size available for that period from 24 to 46 observations.

Period	Start	End	n	Crisis Type
Full Sample	Jan 2000	Dec 2025	312	Baseline
Dot-Com Crash	Mar 2000	Oct 2002	32	Tech bubble / equity selloff
Great Recession	Jun 2006	Jun 2009	37	Housing crisis / credit crisis
COVID-19	Mar 2020	Feb 2022	24	Pandemic / fiscal response
Geopolitical Tensions	Mar 2022	Dec 2025	46	Wars / trade policy

Table 1: Sample period definitions and observation counts.

C. Statistical Method

This paper uses Pearson's correlation coefficient as the primary statistic for the full sample, and reports Spearman's rank correlation coefficient for each of the four crisis sub-periods. The rationale for this dual approach is due to the sample size. For the full sample ($n = 312$), the large number of observations provides a reasonable basis for the use of Pearson's correlation. The Central Limit Theorem suggests that with a sample size this large, the sampling distribution of relevant statistics approaches normality. Pearson's correlation is therefore used as the primary statistic for the full sample.

For the four crisis sub-periods, sample sizes are considerably smaller, ranging from 24 to 46 observations. At these sample sizes, normality assumptions are harder to verify, and individual extreme observations can greatly influence Pearson's coefficient. For this reason, Spearman's rank correlation coefficient is reported. Spearman's ρ (rho) is calculated by converting both sets of data into ranks within each sub-period and calculating the Pearson correlation of those ranks.

Statistical significance for both coefficients is assessed using a two-tailed t-test, with degrees of freedom (DF) equal to n minus 2 in each case. The null hypothesis is that the true correlation is equal to zero. A p-value below 0.05 is the threshold for rejecting the null hypothesis at the conventional 5 percent significance level.

IV. Results

A. Full Sample

Table 2 and Figure 1 present the Pearson correlation result and the corresponding scatter plot for the full 312-month sample.

Period	n	Pearson r	T-Statistic	DF	p-value (2-tail)
Full Sample	312	0.0587	1.0346	310	0.3017

Table 2: Pearson correlation result, full sample, January 2000 to December 2025.

The full-sample Pearson correlation is small and not statistically significant ($r = 0.0587$, $p = 0.3017$). This indicates no meaningful average linear relationship between gold and S&P 500 monthly returns over the 26-year period.

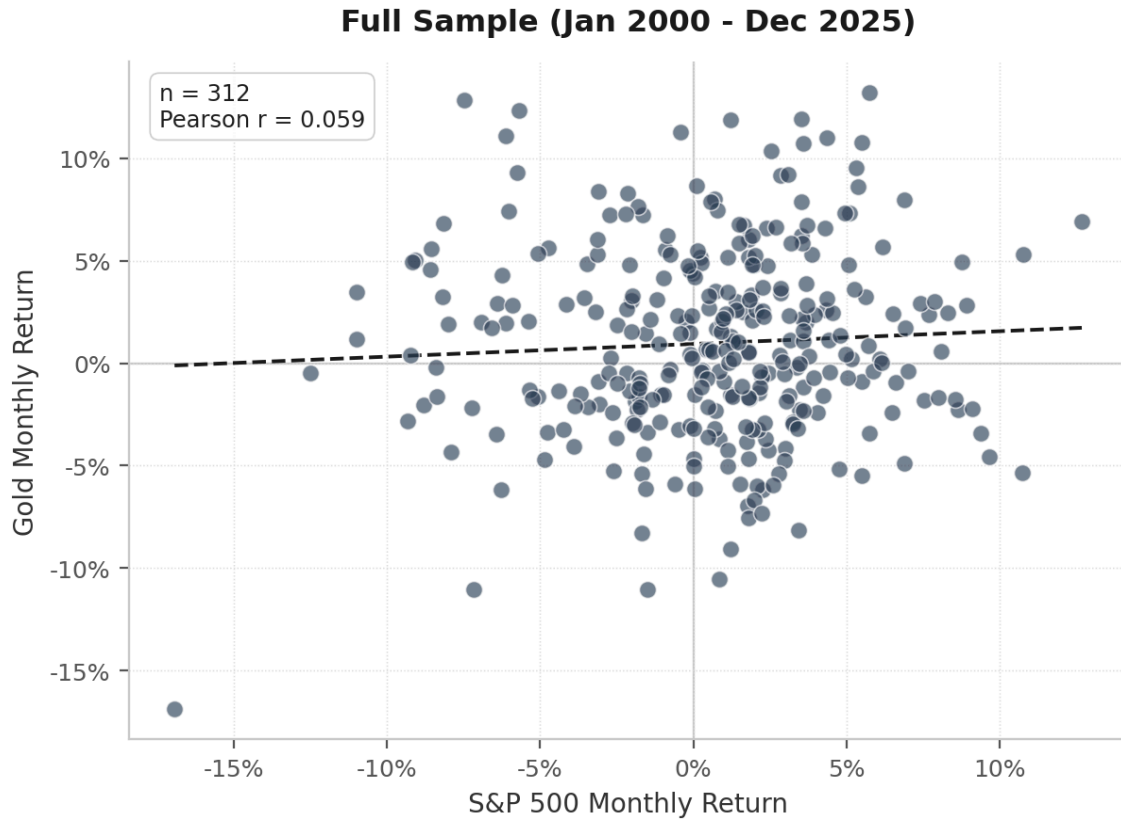


Figure 1: Scatter plot of gold monthly returns against S&P 500 monthly returns, full sample, January 2000 to December 2025. The dashed line is the ordinary least squares fit.

B. Crisis Sub-Periods

Table 3 presents Spearman's ρ for each of the four crisis sub-periods. Figures 2 through 5 present the corresponding scatter plots.

Period	Spearman ρ	DF	p (Spearman)
Dot-Com Crash	-0.1749	30	0.3385
Great Recession	-0.1631	35	0.3348
COVID-19	0.3374	22	0.1069
Geopolitical Tensions/Tariffs	0.1169	44	0.4392

Table 3: Pearson and Spearman correlation results for the four crisis sub-periods. p -values are two-tailed.

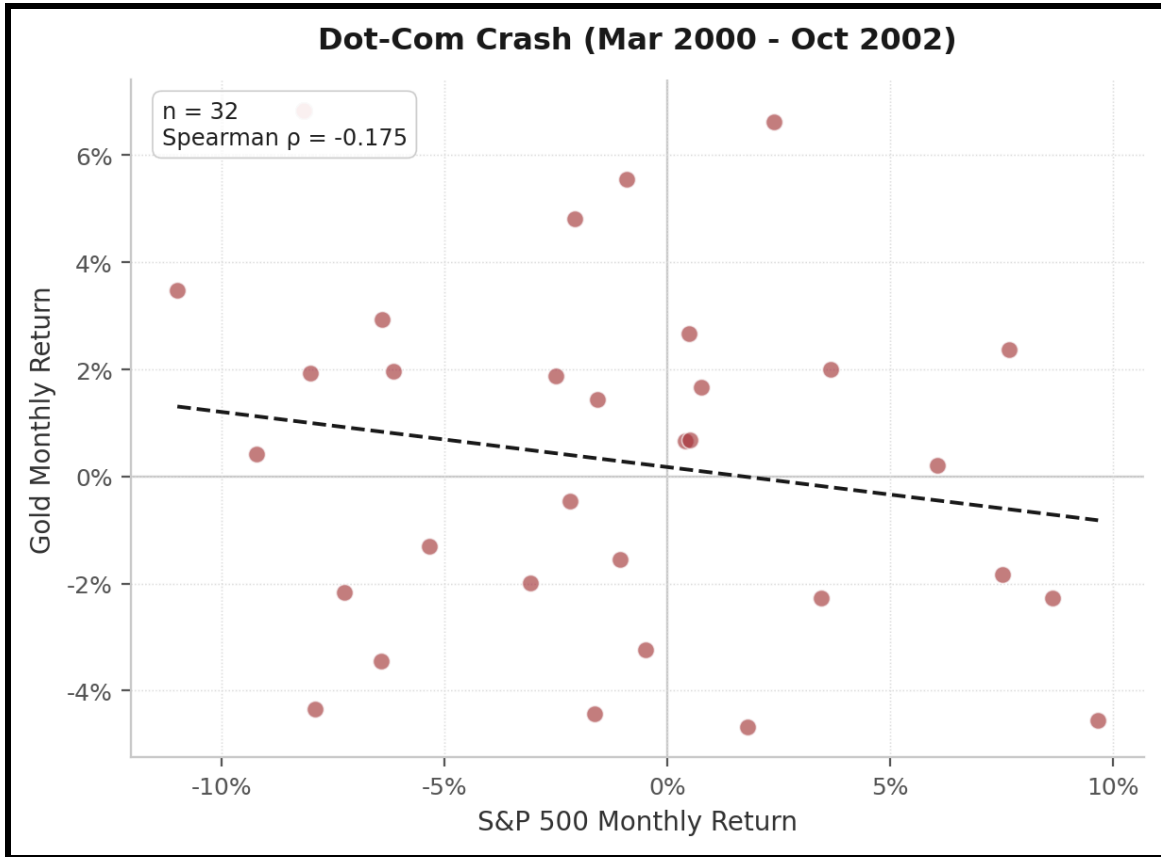


Figure 2: Scatter plot of gold monthly returns against S&P 500 monthly returns, Dot-Com Crash, March 2000 to October 2002.

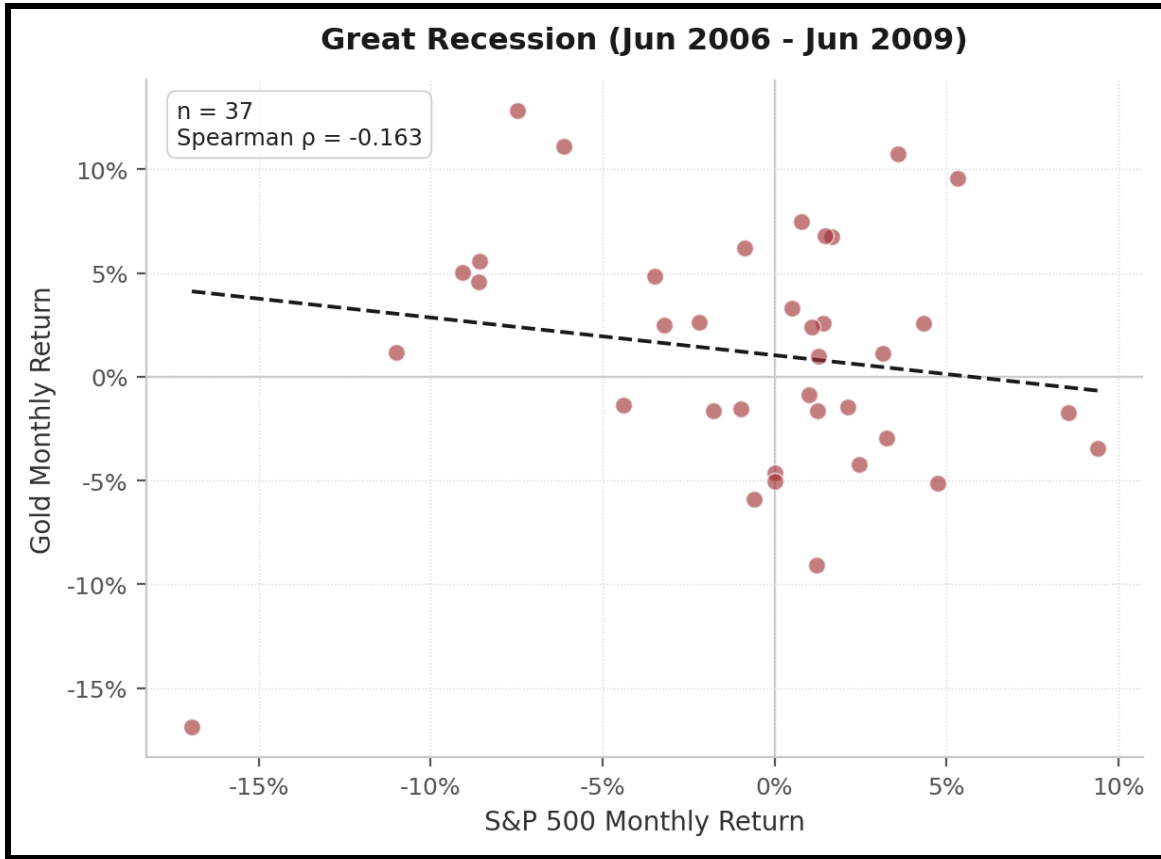


Figure 3: Scatter plot of gold monthly returns against S&P 500 monthly returns, Great Recession, June 2006 to June 2009.

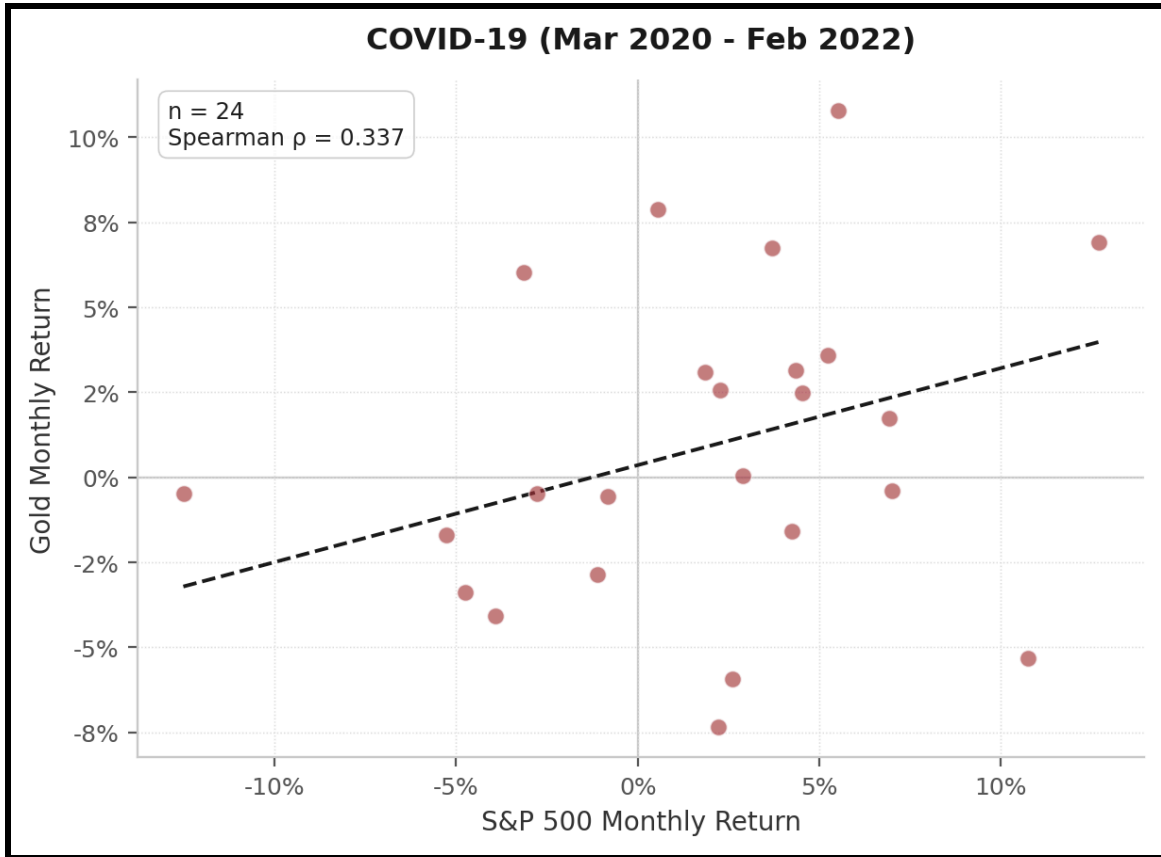


Figure 4: Scatter plot of gold monthly returns against S&P 500 monthly returns, COVID-19, March 2020 to February 2022.

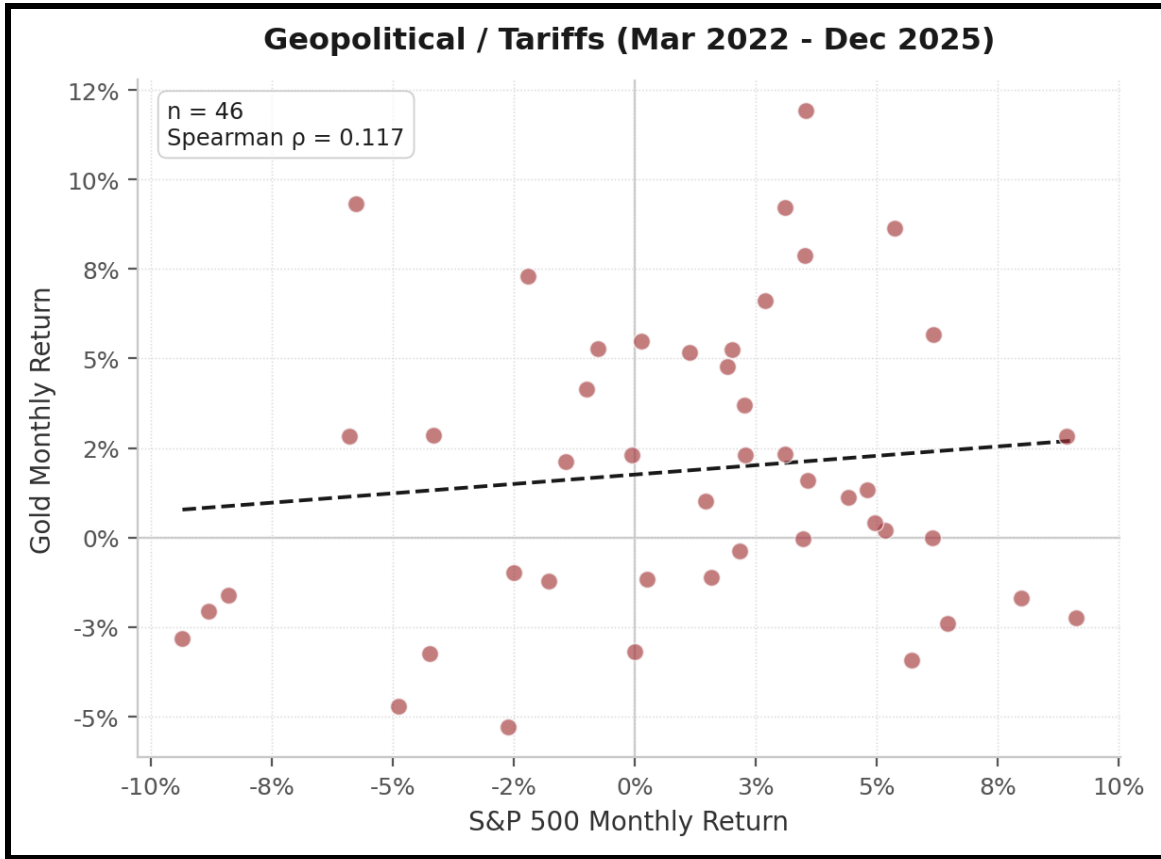


Figure 5: Scatter plot of gold monthly returns against S&P 500 monthly returns, Geopolitical and Tariffs period, March 2022 to December 2025.

Although none of the estimated correlation coefficients are statistically significant at the 5% level, the results still provide useful economic insight. Statistical insignificance only indicates that the observed correlations are not reliably different from zero in a strict statistical sense, but it does not imply the absence of meaningful variation across market environments.

V. Discussion

A. Full Sample

The near-zero full-sample Pearson correlation ($r = 0.0587$) implies that there is little to no relationship between gold and S&P 500. When using the Baur and Lucey framework, a hedge requires a negative or zero average correlation, while a diversifier requires a reliably positive one (Baur and Lucey 2010, 219). Since the coefficient suggests extremely weak correlation, the results could suggest that gold is consistent with the behavior as a weak hedge against equities.

B. Dot-Com Crash and Great Recession

The negative correlations observed during the Dot-Com Crash (Spearman $\rho = -0.1749$) and the Great Recession (Spearman $\rho = -0.1631$) are directionally consistent with the safe-haven definition under the Spearman measure. In both periods, the relationship suggests that as equity markets declined, gold provided at least some safety.

During both the Dot-Com Bubble and the Global Financial Crisis, investors faced increasing uncertainty about the economy and corporate earnings. As confidence in equities weakened, many investors reduced their exposure to stocks and shifted toward assets perceived as safer. Gold has long been viewed as a store of value because it has no default risk and is not directly tied to the financial performance of companies. This increase in demand for gold, alongside reduced demand for equities, is consistent with the weak negative correlations observed during these periods (-0.1749 and -0.1631 , respectively). Although these correlations are not statistically significant, their direction suggests that gold tended to move opposite the stock market during these crises. Under the framework of Baur and Lucey, this behavior is consistent with, but does not conclusively demonstrate, safe-haven characteristics (Baur & Lucey 2010, 219).

C. COVID-19

The COVID-19 period produces the strongest and most directionally surprising result in the analysis. The correlation Spearman ($\rho = 0.3374$) points to a positive relationship. This contradicts the safe-haven prediction. During this period, gold and S&P 500 returns tended to move in the same direction rather than opposite directions, as shown by the upward-sloping cluster of points in Figure 4.

This result is consistent with the early stages of the COVID-19 crisis in March 2020, when investors sold both gold and equities to raise cash during a period of extreme market uncertainty. Rather than moving in opposite directions, both assets declined together before the Federal Reserve's policy interventions helped stabilize financial markets. As markets recovered from April 2020 onward, gold and the S&P 500 also rose together, contributing to the positive correlation observed over the full period. Under Baur and Lucey's framework, this suggests that gold did not act as a safe haven during the most severe phase of the COVID-19 market downturn (Baur and Lucey 2010, 219).

D. Geopolitical Tensions and Tariffs

The most recent period, spanning the Russia-Ukraine war, the Israel-Hamas conflict, and the Trump tariff announcements of 2025, produces a weak but positive correlation (Spearman $\rho = 0.1169$). Figure 5 shows a loosely scattered cloud of points with only a mild upward tilt. While it is slightly weaker than the other correlations observed during the different crises, this positive directional signal suggests gold and equities moved together during this period, indicating some co-movement. This period is characterized by multiple overlapping drivers, including geopolitical risk, which traditionally supports gold prices, inflation uncertainty, rising real interest rates, and trade policy volatility. The mixture of these conflicting forces may explain why the positive correlation, though present, remains weak.

E. Classifying Gold Using the Baur and Lucey Framework

Applying the definitional framework from Baur and Lucey to the full set of results, the evidence does not support classifying gold as a definite hedge, diversifier, or safe haven over this sample period, but the lack of definite correlation over time can suggest that gold has characteristics of a weak hedge (Baur and Lucey 2010, 219). The crisis-period correlations are negative during the Dot-Com Crash and Great Recession, which is consistent with the behavior of a safe-haven, but positive during COVID-19 as well as the geopolitical tensions, which contradicts the safe-haven definition. Gold's role appears to be dependent on the type of crisis rather than structurally defined.

VI. Limitations

There are some limitations applied to this analysis. First, the lack of sample size, especially during the sup-periods, limited the statistical power for all five tests. The inability to reject the null hypothesis cannot be interpreted as evidence that no relationship exists. However, the data may be insufficient to detect a real but moderate correlation. Future research using weekly or daily return data would substantially increase observation counts within each crisis window.

Secondly, this analysis is limited to a bivariate relationship between gold and the S&P 500. Gold prices are known to be influenced by a range of macroeconomic variables including real interest rates, the U.S. dollar exchange rate, and inflation expectations. Excluding these variables means that the measured correlation may reflect confounding factors rather than a direct relationship between the two assets.

Thirdly, the extension of the Great Recession period back to June 2006 introduces some judgment regarding when the housing crisis transitioned into a systemic financial crisis. An earlier or later start date would change the composition and result of this sub-period.

VII. Conclusion

This paper examines whether gold functions as a hedge, diversifier, or safe haven relative to the S&P 500 using monthly return data from January 2000 to December 2025. Pearson's correlation is used as the primary statistic for the full 312-month sample, supported by the large sample size. For the four crisis sub-periods, Spearman's ρ is used since smaller samples are more vulnerable to distortion from non-normality and individual extreme observations, making it the preferred correlation coefficient to measure.

No statistically significant correlation is found in any period under either method at the 5 percent level. However, the directional pattern of the coefficients provides meaningful economic insight. When observing the whole time frame, the little to no correlation between gold and S&P 500 suggests that gold has characteristics of a weak hedge according to Baur and Lucey's framework (Baur and Lucey 2010, 219). During the Dot-Com Crash and the Great Recession, correlations are negative under the Spearman measure, directionally consistent with safe-haven behavior even if not statistically conclusive. However, during the COVID-19, the correlation was positive, and during the most recent geopolitical period, the correlation coefficient suggests a weak positive correlation. When applying Baur and Lucey's definitions, the gold-equity relationship appears to be regime-dependent, shaped by the specific characteristics of each crisis rather than a stable structural relationship between the two asset classes (Baur and Lucey 2010, 219).

For investors, these findings suggest that gold should not be viewed as a consistently reliable hedge, safe haven, or diversifier against equity markets. The results support a regime-dependent interpretation in which gold's effectiveness changes with the underlying source of the crisis. Future research using higher-frequency data, as well as incorporating additional variables such as interest rates and the U.S. dollar, and refining the crisis periods, would provide a more complete understanding of how gold behaves.

VIII. References

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